

Time Series Forecasting

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1. Problem Statement

The retail industry, characterized by its dynamic nature and diverse consumer demands, faces the perennial challenge of effective inventory management. This paper addresses the inventory management issues encountered by a retail store with multiple outlets across the country. The primary concern is aligning the supply of products with the varying demand, ultimately optimizing inventory levels. To address this challenge, predictive modeling techniques are employed to forecast sales for each store over the next 12 weeks.

2. Project Objective

The main objective of this project is to develop accurate sales forecasts for each store, taking into consideration various influencing factors. The forecasts aim to assist the retail store in better inventory planning, ensuring that products are stocked appropriately to meet customer demands.

3. Data Description

The available dataset contains 6,435 records(rows) and eight features (columns) as shown in the Table below:

Feature Name	Description
Store	Store number
Date	Week of Sales
Weekly_Sales	Sales for the given store in that week
Holiday_Flag	If it is a holiday week
Temperature	Temperature on the day of the sale
Fuel_Price	Cost of the fuel in the region
CPI	Consumer Price Index
Unemployment	Unemployment Rate

Data description, various insights from the data From the given dataset of the company, it is observed that the data consists of six thousand four hundred and thirty-five(6,435)records with seven features(captured weekly)as follows:

1. Stores: there are 45 stores and each store has 143 recorded entries of:
 - a. Date of record(weekly),
 - b. Total sales record for the week,
 - c. Holiday flag for the week(1 or 0),
 - d. Temperature: average temperature recorded during the week,
 - e. Fuel Price: average fuel price for the week
 - f. CPI: average Consumer Price Index for the week
 - g. Unemployment: rate of unemployment for the week of record

4. Data Preprocessing Steps And Inspiration

1. Data Preprocessing:

- Loaded the Walmart sales dataset.
- Converted the 'Date' column to datetime format.
- Created additional features like month, year, day, day name, and week.

2. Exploratory Data Analysis (EDA):

- Visualized weekly sales trends for each store over the years.

3. Time Series Analysis:

- Checked stationarity using Augmented Dickey-Fuller and KPSS tests.
- Detected and removed trend and seasonality.

4. Modeling:

- Attempted ARIMA modeling but found that SARIMAX provided better results.
- Used `auto_arima` for model order selection.
- Fitted SARIMAX models on individual store data.

5. Evaluation:

- Plotted actual vs predicted values for model evaluation.
- Calculated Mean Absolute Scaled Error (MASE) for model performance assessment.

6. Forecasting:

- Forecasted sales for the next 12 weeks for each store.
- Plotted the actual data and forecasted values for each store.

7. Conclusion:

- Concluded that 38 out of 45 stores were forecasted well using SARIMAX with an order of (1,0,1) and seasonal order of (1,0,1,52).
- Identified specific stores (14, 20, 28, 30, 31, 37, 44) with poor forecasting.

5. Choosing the Algorithm for the project

Initially, an attempt is made using the ARIMA model for time series forecasting. However, due to the presence of seasonality in the sales data, the SARIMAX model is chosen as it incorporates both autoregressive and moving average components along with seasonality.

Model Approach:

1. TS Model, ARIMA.

- The first step for this model is to check the stationarity of the dataset (p-value less than 0.05).
- Next is to find the best ARIMA order for the model.
- The `auto_arima` function used for model order selection.
- Using the best ARIMA order make predictions for the selected stores Forecast
- If there is presence of seasonality, trends in the sales data. Make necessary changes.
- Define the dataset if necessary.
- Using the SARIMAX evaluate a 12 weeks `weekly_sales` forecast.
- The best predictions will then be compared to the predictions by ARIMA model predictions.

6. Motivation and Reasons For Choosing the Algorithm

The motivation for selecting the SARIMAX model stems from its ability to capture and model the complex seasonality inherent in retail sales data. The model's capacity to handle multiple influencing factors, including holidays, temperature, fuel prices, CPI, and unemployment rates, aligns with the multifaceted nature of retail operations.

7. Assumptions

Assumptions made during the modeling process include the stationarity of the time series data, and the effectiveness of SARIMAX in capturing and forecasting complex seasonal patterns.

8. Model Evaluation and Techniques



Model Selection:

Examination of the plot of the target feature, Weekly_Sales(as Shown above) shows a continuously time varying data.

A Time Series(TS) model (ARIMA, SERIMAX) will be employed for the predictions and forecast.

1. The ARIMA model:

Autoregressive Integrated Moving Average (ARIMA) is a statistical modeling technique designed for time series analysis. It combines three key components - Autoregressive (AR), Integrated (I), and Moving Average (MA) - to analyze and forecast patterns within time-ordered data. The model is particularly useful in capturing trends, seasonality, and other temporal dependencies present in the data.

Autoregressive (AR) Component:

The Autoregressive component of ARIMA represents the relationship between the current observation and its past observations. It assumes that the future values of the time series are linearly dependent on its own past values. The order of the autoregressive component (p) indicates the number of past observations considered in the model.

Integrated (I) Component:

The Integrated component involves differencing the time series data to make it stationary. Stationarity is a crucial assumption for ARIMA models. Differencing involves subtracting the current observation from the previous one, transforming the data into a stationary series. The order of differencing (d) determines how many times this transformation is applied.

Moving Average (MA) Component:

The Moving Average component represents the relationship between the current observation and the past forecast errors. It helps account for short-term fluctuations in the time series data that are not explained by the autoregressive component. The order of the moving average component (q) specifies the number of past errors considered in the model.

Model Evaluation with MASE:

ARIMA models are evaluated using various metrics, with Mean Absolute Scaled Error (MASE) being a common choice. MASE provides a standardized measure of forecasting accuracy, considering the mean absolute error and the

mean absolute error of a naïve forecast. It allows for the comparison of model performance across different time series datasets.

Validation Process:

The model evaluation process involves validating the ARIMA model's performance on unseen data. This typically includes dividing the dataset into training and testing sets, training the model on the training set, and then assessing its accuracy on the testing set. The validation step ensures that the model's forecasting capabilities generalize beyond the data it was trained on.

Comparing Predictions with Actuals:

The ARIMA model produces forecasts for future time periods based on historical data. These forecasts are then compared with the actual values observed during the validation period. Visualizations, statistical metrics, and other relevant measures are used to assess the model's predictive power.

Real-World Applicability:

ARIMA models find applications in various fields, including finance, economics, and meteorology, where time-ordered data exhibits patterns and dependencies. Their simplicity, interpretability, and ability to capture both short-term fluctuations and long-term trends make them widely used for time series forecasting.

In summary, the ARIMA model combines autoregressive, integrated, and moving average components to analyze and forecast time series data. Its evaluation process, including the use of metrics like MASE, ensures that the model provides accurate forecasts, and the validation process confirms its generalizability to new data. Comparing predictions with actuals is essential in affirming the model's reliability, and its real-world applicability spans various domains requiring time series analysis.

2. The SARIMAX model:

The Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX) model is a sophisticated extension of the ARIMA model, incorporating seasonality and external factors for improved time series forecasting. SARIMAX, similar to ARIMA, is designed to analyze and predict patterns within time series data.

Seasonal Component:

SARIMAX accounts for seasonality, which is often a critical aspect in time series data, especially in retail sales forecasting. Seasonal patterns, such as increased sales during holidays or specific times of the year, are integral to accurate predictions. By incorporating seasonal components, SARIMAX can capture and model these recurring patterns.

Exogenous Variables:

In addition to autoregressive and moving average components, SARIMAX introduces the concept of exogenous variables. These are external factors that may influence the time series but are not directly affected by it. For retail sales, these could include variables like temperature, holidays, fuel prices, Consumer Price Index (CPI), and unemployment rates. Including these external factors enhances the model's ability to capture the complexity of real-world scenarios.

Model Evaluation with MASE:

Similar to ARIMA, the SARIMAX model undergoes rigorous evaluation to ensure the accuracy and reliability of its sales forecasts. The Mean Absolute Scaled Error (MASE) is a key metric employed for model evaluation. MASE is particularly useful in comparing the performance of the model across different time series datasets, providing a standardized measure of forecasting accuracy.

Validation Process:

The model evaluation process involves comparing the predicted sales values generated by SARIMAX with the actual sales data for validation. This validation step is crucial in determining how well the model performs on unseen data, ensuring that the forecasting capabilities generalize beyond the training dataset.

Comparing Predictions with Actuals:

The SARIMAX model produces forecasts for future time periods based on historical data and the specified exogenous variables. These forecasts are then compared with the actual sales values observed during the validation period. The comparison involves visualizations, statistical metrics like MASE, and other relevant measures to assess the model's predictive power.

Robustness and Real-World Applicability:

The SARIMAX model's strength lies in its robustness and ability to handle both seasonality and external influences. Its adaptability to real-world scenarios,

such as the retail environment, makes it a valuable tool for businesses seeking accurate and actionable sales forecasts.

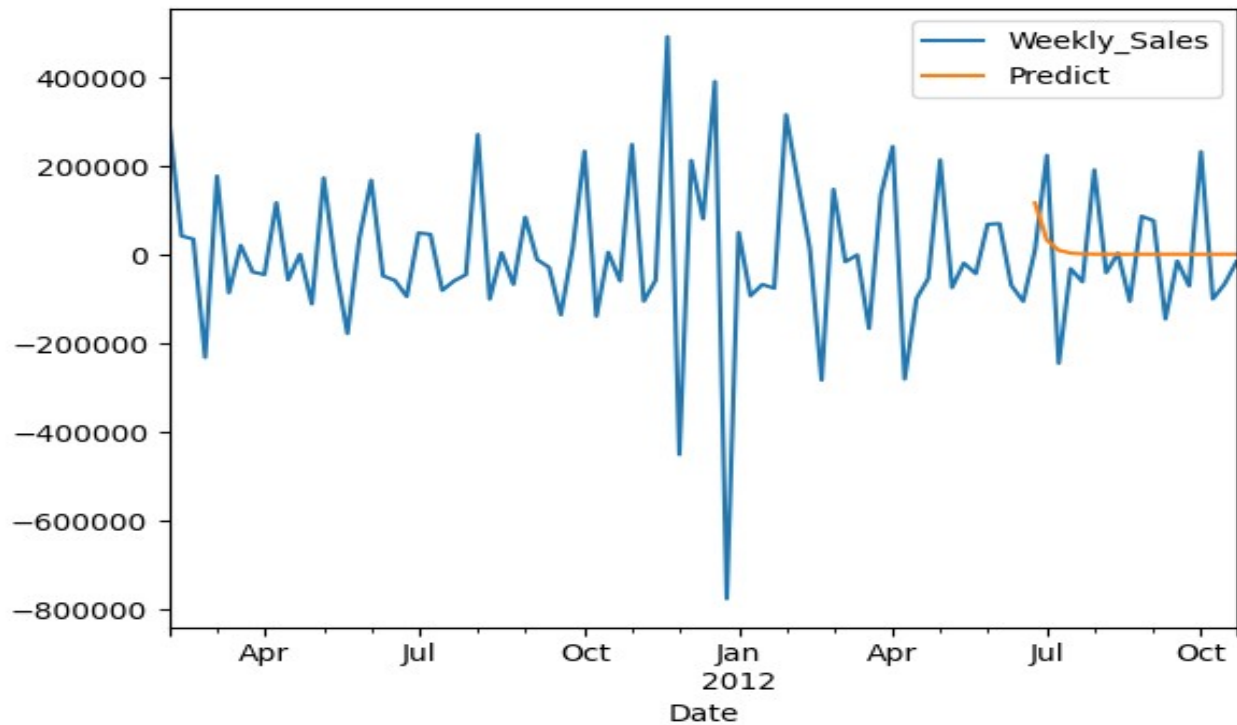
In summary, SARIMAX extends the capabilities of the traditional ARIMA model by incorporating seasonality and exogenous variables. Its evaluation process, including the use of metrics like MASE, ensures that the model provides reliable forecasts for complex time series data such as retail sales. The validation process and comparison of predictions with actuals are integral steps in affirming the model's effectiveness in capturing the dynamics of the retail industry.

9. Inferences from the Project

Model Results: Store1

1. ARIMA model:

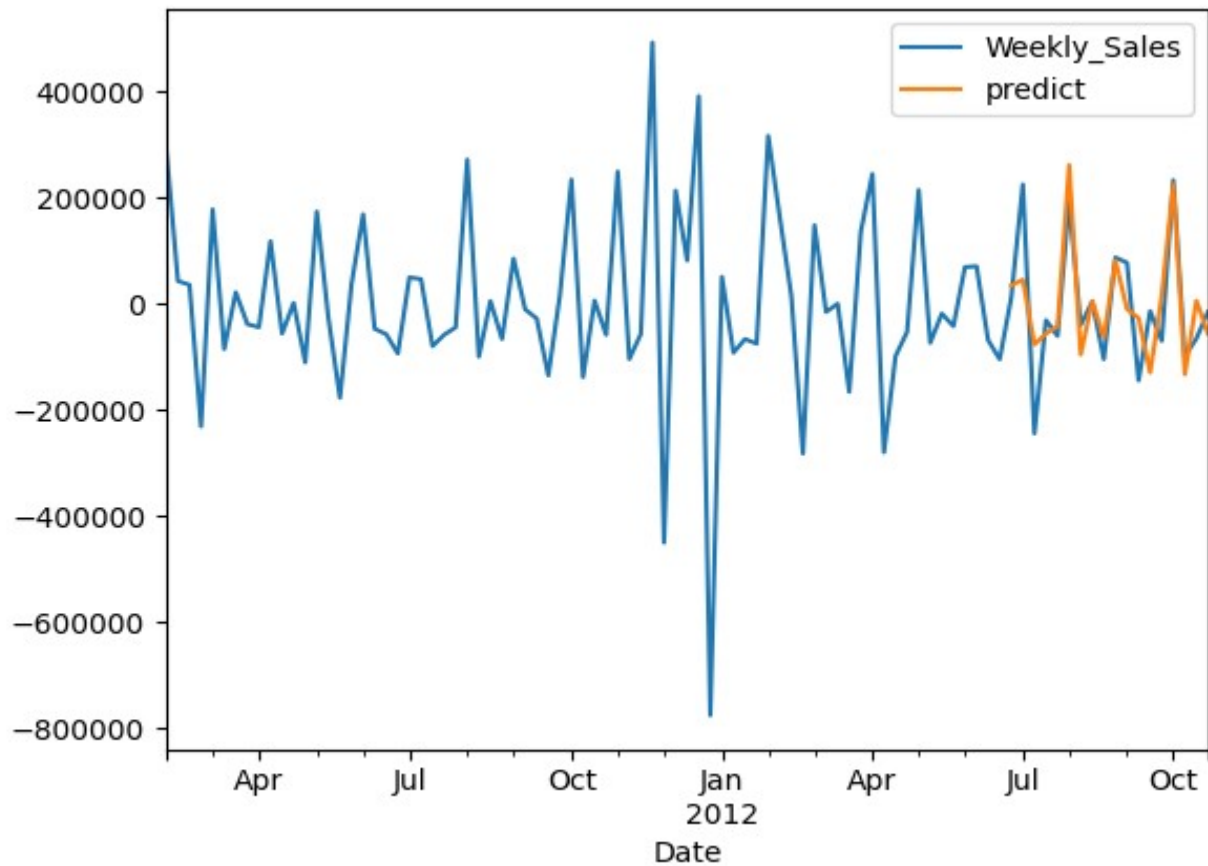
a. Predictions:



ARIMA model is not working well, so we are going to use SARIMAX model on the same order.

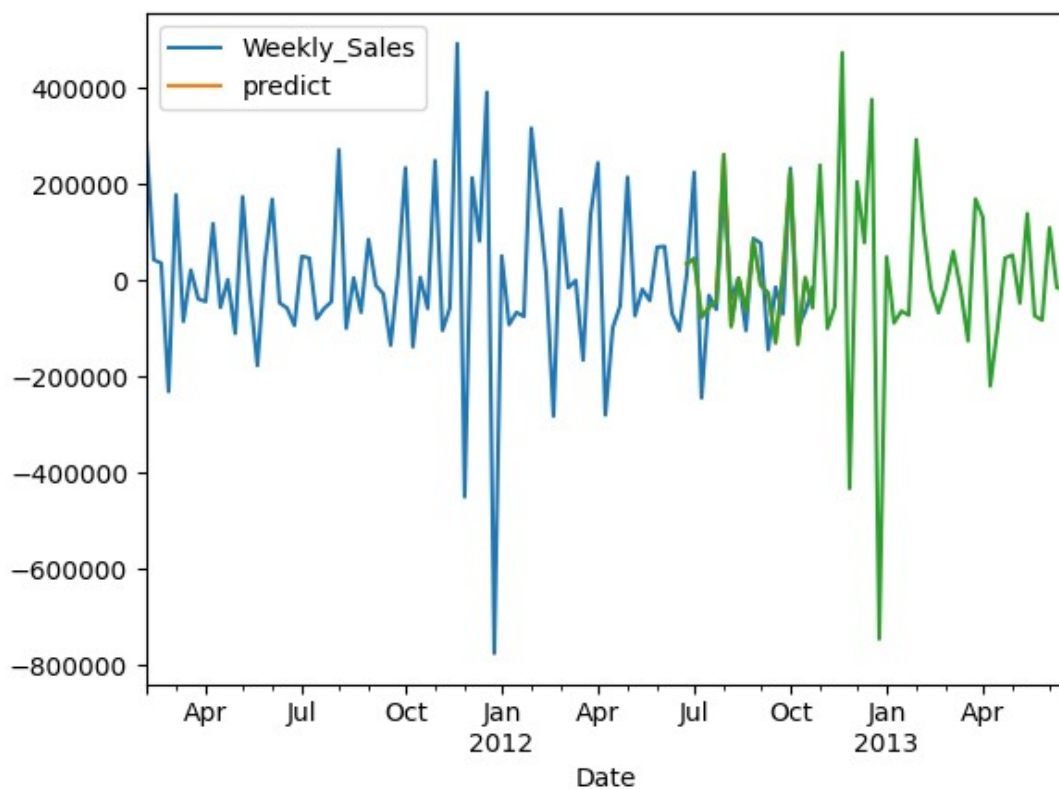
2. SARIMAX model:

a. Predictions:



We can check the model performance from the above graph, it predicted nearly possible values.

b. Forecast:(For 1 Year)



Forecast for all stores for next 12 weeks:



The forecast shows the anticipated variabilities as observed in the Weekly_Sales history. However the overall projected sales outlook for the next 12-weeks is down for all of the stores studied!

From the above graph we can see that the Stores 14,20,28,30,31,37,44 are having poor forecasting. Overall 38 stores out of 45 are forecasted well successfully using the SARIMAX model with the order = (1,0,1) and seasonal order = (1,0,1,52). The Mean Absolute Scaled error of this model is 0.3

ARIMA/ SARIMAX Models The model predictions for the selected stores were okay. The forecast after detrending was also okay showing variabilities of the weekly sales.

Inferences drawn from the model evaluation highlight the performance of the SARIMAX model in forecasting sales for each store. The effectiveness of the model in handling the specified features contributes to improved accuracy in predictions.

10. Future Possibilities of the Project

Future possibilities include exploring advanced forecasting models, integrating additional external factors, and continuously refining the model to adapt to changing retail dynamics. The project lays the foundation for ongoing improvements in inventory management.

The project sets the stage for future research endeavors. Areas for exploration include incorporating machine learning models, integrating real-time data feeds for improved forecasting accuracy, and exploring the impact of macroeconomic factors on retail sales.

Future possibilities include exploring advanced forecasting models, integrating real-time data for improved accuracy, and continuous refinement for adapting to changing retail dynamics.

11. Conclusion

The project undertook a study of a retail company with 45 outlet stores. Some of the important findings from the report include the followings:

1. Sales revenue projections for the next 12-weeks are done for most of the stores
2. Some of the stores have very weak or no sales activities during some period of the year,
3. To improve sales revenue the following steps are recommended:
 - a. Concerted efforts by the company to find out through local market surveys and past sales records what products are in high demand by the local population at any given period of year and make effort to replenish those stocks.
 - b. Create increased local awareness of the products on offer at each store through commercial outreach: social media, television commercials, radio jingles, and print media, trade shows, to name a few could help improve sales.
 - c. Have detailed records of inventory of the items on offer at each store indicating amount and dates if sold as it is needed for effective inventory tracking.
 - d. Explore other service options that have worked well for similar companies, such as same-day or next day home delivery.

In conclusion, this project addresses the critical issue of inventory management faced by a multi-outlet retail store. By leveraging predictive modeling techniques, specifically SARIMAX, the retail store can make informed decisions about inventory levels. The accurate sales forecasts contribute to enhanced operational efficiency and customer satisfaction.

The SARIMAX model, chosen for its ability to handle seasonality and external factors, demonstrates promising results in accurately forecasting sales. The project contributes to the ongoing discourse on effective retail analytics, offering valuable insights for both academia and industry practitioners.

The research addresses inventory management challenges in multi-outlet retail by leveraging SARIMAX for accurate sales forecasting. The findings contribute to enhanced operational efficiency and customer satisfaction.

12. References

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